NO.1 i-stop WARNING LIGHT (AMBER) ILLUMINATES [SKYACTIV-D 1.5]

id1103q4000600

1	i-stop WARNING LIGHT (AMBER) ILLUMINATES
DESCRIPTION	• i-stop warning light (amber) remains illuminated while engine is running.
POSSIBLE CAUSE	**Note** • After switching the ignition ON, the i-stop warning light (amber) turns off after illuminating for 1.5 s for a burnt out bulb check (at engine start). **System off recognized due to false detection of i-stop OFF switch** • i-stop OFF signal malfunction — i-stop OFF switch malfunction — Short to ground in wiring harness between cluster switch terminal B and instrument cluster terminal 1V **Instrument cluster malfunction** • i-stop warming light (amber) illumination circuit in instrument cluster disabled • CAN communication line malfunction between instrument cluster and PCM

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**VERIFY DTC** • Retrieve the PCM and instrument cluster DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	Go to the next step.
2	**DETERMINE FALSE ILLUMINATION OF i-stop WARNING LIGHT (AMBER)** • Does the i-stop operate while the i-stop warning light (amber) is illuminated?	Yes	False illumination of i-stop warning light (amber) can be considered the cause. Go to the next step.
		No	Go to Step 4.
3	**DETERMINE IF MALFUNCTION CAUSE IS INSTRUMENT CLUSTER WARNING LIGHT ILLUMINATION CIRCUIT OR ERROR SIGNAL FROM PCM** • Switch the ignition ON (engine off). • Access the instrument cluster PID WL+IL and illuminate and turn off the warning light in the instrument cluster using the M-MDS. (See ACTIVE COMMAND MODES INSPECTION [INSTRUMENT CLUSTER].) • Does the i-stop warning light (amber) illuminate or turn off according to the M-MDS operation? (See ACTIVE COMMAND MODES TABLE [INSTRUMENT CLUSTER].)	Yes	Repeat the inspection from Step 1. • If the malfunction is not resolved, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 6.
		No	Replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.) Go to the next step.
4	**INSPECT i-stop OFF SWITCH** • Inspect the i-stop OFF switch. (See i-stop OFF SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the cluster switch. (See CLUSTER SWITCH REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5*	**DETERMINE IF MALFUNCTION CAUSE IS i-stop OFF SWITCH SIGNAL OR OTHER** • Switch the ignition off. • Disconnect the cluster switch and instrument cluster connectors. • Inspect for continuity between cluster switch terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between cluster switch terminal B and instrument cluster terminal 1V. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Replace the instrument cluster, then go to the next step. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)
6	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		